

Uzair Shafique

Data Scientist & Machine Learning Engineer

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SUMMARY

Data Scientist and Machine Learning Engineer with experience building end-to-end machine learning and AI projects from problem definition to deployment. Skilled in data preparation, model development, backend development, API design, interactive web applications, and deployment using Python and machine learning libraries and frameworks. Experienced in applying sound software engineering practices to develop practical, maintainable, and user-focused projects.

PROFESSIONAL EXPERIENCE

DevOps & Cloud Intern

10/2024 – 12/2024

Al Nafi Global Internship Program

Remote

- Completed practical training in Linux, Docker, Kubernetes, cloud computing, and DevOps fundamentals.
- Worked with Linux command-line tools, file systems, permissions, and documentation practices.
- Gained hands-on experience in GRC, containerization, SysOps, blockchain fundamentals, and cloud technologies.

Machine Learning Engineer

06/2025 – 12/2025

(Independent Client) Project: Medicare Part D Drug Spending Forecasting Tool

Remote

- Collaborated with the client to define forecasting objectives, requirements, and project scope.
- Developed an end-to-end machine learning pipeline using LightGBM and Optuna, covering data preparation, feature engineering, and model evaluation.
- Built an interactive Streamlit dashboard and organized a modular, maintainable codebase with deployment-oriented practices.

Business Data Analyst Intern

02/2023 – 08/2023

Forage (Virtual Internship)

Remote

- Analyzed and prepared business datasets to support data-driven decision-making.
- Created visualizations to communicate insights clearly and effectively.
- Applied analytical and problem-solving skills through practical business case studies.

EDUCATION

Doctor of Pharmacy (Pharm.D)

09/2020 – 04/2026

Islamia University of Bahawalpur (IUB)

Bahawalpur, Pakistan

- Studied pharmacology, medicinal chemistry, pharmaceuticals, pharmacotherapy, clinical pharmacy, pharmaceutical analysis, microbiology, pathology, and patient care as part of the Pharm.D curriculum.
- Developed an interest in machine learning and computational approaches alongside the Pharm.D curriculum.
- Applied pharmaceutical knowledge with AI and machine learning by building projects such as a QSAR prediction system for molecular bioactivity prediction.

PROJECTS

QSAR Prediction using Machine Learning 🔗

Computational Drug Discovery | Python | MLOps

- Built an end-to-end QSAR Streamlit app to predict molecular bioactivity from SMILES using machine learning and molecular descriptors.
- Developed a FastAPI backend and Streamlit frontend with REST APIs and SHAP-based model explanations.
- Containerized the application with Docker and deployed it as an interactive web application.

StudyMind AI (RAG Study Assistant)

Python | LangChain | Ollama

- Developed a Retrieval-Augmented Generation (RAG) study assistant for document search, question answering, and quiz generation.
- Implemented local and cloud inference workflows using FAISS vector search and large language models.
- Designed an intuitive Streamlit interface for document upload and interactive learning.

Nexus IT Assistant

Python | LangGraph | LangChain | Ollama

- Built an AI-powered IT support assistant Streamlit app using the ReAct agent pattern with LangGraph.
- Implemented automatic switching between local Ollama models and cloud-based models.
- Developed a conversational interface with tool calling and memory for interactive support workflows.

DocScan OCR Pipeline

Python | MLOps | Docker

- Engineered a production-oriented OCR pipeline using EasyOCR with FastAPI and Streamlit.
- Integrated MLflow for experiment tracking, Docker for deployment, and monitoring for model reliability.
- Designed a modular architecture to support scalable OCR workflows.

SKILLS

Programming: Python | SQL

Machine Learning: Scikit-learn | XGBoost | LightGBM | TensorFlow | PyTorch

Generative AI: LangChain | LangGraph | RAG | Ollama | Gemini | Prompt Engineering

Backend: FastAPI | Flask | REST APIs

Frontend: Streamlit | React | HTML | CSS

Data: Pandas | NumPy | SQLite | Excel

MLOps and Deployment: Docker | Git | GitHub | AWS | Render

Visualization: Plotly | Matplotlib

PUBLICATIONS

AI Insights & Innovations Newsletter

LinkedIn

- Publish technical articles on machine learning, generative AI, deployment, MLOps, and practical software engineering.
- Share lessons from building end-to-end AI/ML projects, with a focus on solving real-world problems and production-oriented development.
- Built a community of **1,000+ subscribers** on LinkedIn.

CERTIFICATES

Retrieval Augmented Generation (RAG) with LangChain:  DataCamp

Amazon Web Services Machine Learning Essential Training:  LinkedIn Learning

Intro to ChatGPT and Generative AI:  365 Data Science